

NetEngine 8000 M1A Service Router

Overview

NetEngine 8000 M1A is a high-density compact router designed for the cloud era. It's featured with 1U height, 220 mm depth, and supports up to 176G port capacity. The compact design saves precious space resources. It supports flexible service access of 10GE and GE (optical and electrical ports) and supports features such as SRv6, EVPN, Telemetry, and 1588v2 high-precision clock, which is the best choice for multi-service high-density bearer in the future cloud era. The product fully supports the future-oriented next-generation unified network SRv6 technology and can be used to build IP backbone, metro, mobile backhaul, and data center networks. It can be used for individual/enterprise cloud migration, inter-cloud interconnection, IoT, government and enterprise private lines, home broadband applications, and CDN. As a basic bearer protocol of the entire network, SRv6 implements end-to-end full-service bearer capabilities. SRv6 enables simplified protocols, large-scale networking, seamless integration, high reliability, integrated service chains, network and service programmability, and a full ecosystem.

- Support for data redundancy elimination (DRE) to compress and decompress data between WAN border routers, reducing bandwidth consumption of WAN links, reducing line costs, and accelerating the deployment of distributed data centers.

The NetEngine 8000 series router integrates multiple functions, simplifies the network structure, provides rich service types, reliable service quality, and intelligent O&M. It leads the IP WAN to an intelligent network with self-driving driving, and continuously drives the business success of enterprise customers.



NetEngine 8000 M1A (DC)



NetEngine 8000 M1A (AC)

Product Highlights

Compact and high-densityGE

- volume: 220mm deep, 1U
- capacity : 176 Gbps, Up to 16*10GE/32*GE

SRv6 Ready, Simplified network

- SRv6/EVPN: Simplified protocols
- Programmable NP, meeting future evolution requirements

Energy saving

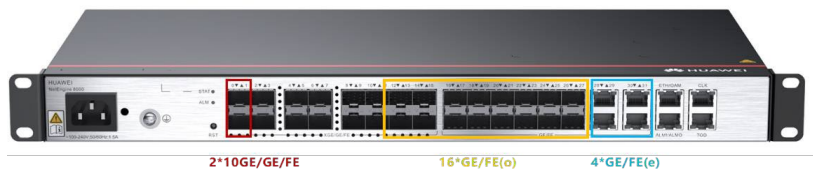
- -40°C ~ 65°C, Industrial wide temperature design
- 80W, Industry-leading low power consumption

Product Specifications

Item	Description
Switching Capacity	352 Gbps
Interface Type	10GE/GE/FE(o) /FE(e)
Main Control Board	Fixed-Box
Dimensions (W x D x H)	442 mm × 220 mm × 44.5 mm (1U)
Typical Power Consumption	89.21W
Temperature	-40°C ~ 65°C
Voltage	AC: 200V to 240V/100V to 127V dual live wires, support 240V HVDC; DC: -48 V/-60 V

Port Mode

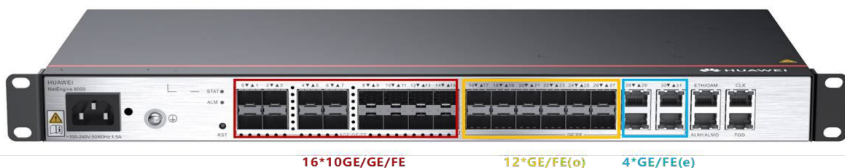
NetEngine 8000 M1A: 4*10G+8*GE(o)+4*GE(e)



NetEngine 8000 M1A: 6*10G+22*GE(o)+4*GE (e)



NetEngine 8000 M1A: 16*10G+12*GE(o)+4*GE (e)



2*10G+8GE by default, Don't need to configure the RTU. The above are typical configurations. You can flexibly specify the default 2*10G+8GE ports.

Product Features

Features	Description
Layer 2	IEEE 802.1q、 IEEE 802.1p、 IEEE 802.3ad、 IEEE 802.1ab、 STP/RSTP/MSTP、 EVC
Layer 3	OSPFv2/v3、 RIPv2、 IS-IS/IS-ISv6、 BGPv4/BGPv4+、 IPv4 ACL/Telnet、 6VPE and Static routing protocol、 Dynamic ARP, static ARP, VLANIF interface and VXLAN
MPLS	LDP、 RSVP-TE、 SR MPLS、 L2VPN (VPLS、 HVPLS、 VLL) 、 L3VPN、 Seamless MPLS、 MPLS-TP

Features	Description
SRv6/EVPN	IS-IS for SRv6、OSPFv3 for SRv6、SRv6 TE Policy、SRv6 TE Policy Shortcut、SRv6 Flex-Algo、SRv6 SRH Compression、SRv6 Network Slicing、G-SRv6、SRv6 BGP、SRv6 BE、SRv6 BE SBFD、SRv6 TI-LFA FRR、SRv6 middle-node protection、SRv6 micro-loop avoidance、SRv6 OAM、SRv6 SFC、EVPN L3VPN、EVPN VPWS、EVPN VPLS
VAS	NAT/CGN、IPSec
Multicast	IGMPv1/v2/v3、IGMP Snooping、Static multicast routing、PIM-SM/SSM、MBGP、NG MVPN、BIERv6
QoS	5-level HQoS、CAR
Reliability	IP FRR、LDP FRR、TE FRR、VPN FRR TE-tunnel APS PW redundancy protection and PW APS Bit error-triggered protection switching
Security	URPF: used to prevent network attacks based on source IP address spoofing. Local attack defense: includes management and control plane protection, attack source tracing, and alarm generation in the event that the discarded packet threshold is crossed. Whitelist association at the application layer IPsec provides security services for IP packets mainly through encryption and authentication. AAA: Authentication, Authorization, and Accounting SSH (Secure Shell) ACL/Layer 2 ACL/ACL6
Maintainability	A network management system (NMS) with a graphical user interface, which simplifies NE management, improves O&M capabilities, and facilitates network-wide or end-to-end performance monitoring and fault diagnosis. Plug-and-play based on DHCP or DCN. The NMS can automatically detect and configure the newly connected devices, which helps to implement remote batch commissioning. Directional Forwarding Detection (BFD), Ethernet OAM, MPLS OAM and MPLS-TP OAM ITU-T Y.1564、ITU-T Y.1731 Bandwidth association with microwave devices is supported. Bandwidth association simplifies QoS configurations and requirements on the microwave device, and the complex QoS logic is implemented on the NetEngine 8000 device. IP FPM、Seamless MPLS SNMPv1/v2c/v3 CLI、NETCONF、RMON

Features	Description
Synchronization	The Device supports complete clock synchronization solutions to provide precise frequency or time synchronization. The following clock features are used: Physical-layer synchronization, including Ethernet clock synchronization Network Time Protocol (NTP) 1588v2, which meets the LTE network's requirements for clock synchronization G8275.1

Regulatory compliance

Item	Description
Regulatory compliance	EMC • ANSI C63.4 • AS/NZSPR 32 • CISPR 24 • CISPR 32 • EN 55024 • EN 55032 • ETSI EN 300 386 • FCC CFR47 Part 15 Subpart B • ICES-003 Issue 6 • ICES-GEN Issue 1 • IEC 61000-3-2 • IEC 61000-3-3 • IEC 61000-4-11 • IEC 61000-4-2 • IEC 61000-4-29 • IEC 61000-4-3 • IEC 61000-4-4 • IEC 61000-4-5 • IEC 61000-4-6 • IEC 61000-6-1 • IEC 61000-6-3 • VCCI-CISPR 32

项目	描述
Regulatory compliance	Environment • ETSI EN 300 019-1-1 • ETSI EN 300 019-1-2 • ETSI EN 300 019-1-3 • ETSI EN 300753 • ETSI EN 300 019-2-1 • ETSI EN 300 019-2-2 • ETSI EN 300 019-2-3 • IEC 60068-2-1 • IEC 60068-2-2 • IEC 60068-2-14 • IEC 60068-2-30 • IEC 60068-2-64 • IEC 60068-2-27 • IEC 60068-2-6 • IEC 60068-2-31 • IEC 60068-2-78 • ISTA 2A • ISO 7779 • GB/T 2423.26-2008 Safety • IEC/EN/UL/CSA 60950-1 • IEC/EN 62368-1 Environmental protection • 2011/65/EU & (EU)2015/863 (EU RoHS) • Regulation (EC) No.1907/2006 (REACH) • 2012/19/EU (WEEE) • 2006/66/EC & 2013/56/EU on batteries and accumulators

NOTE

- Regarding the physical dimensions provided in the table, the width (W) does not take mounting ears into account.
- Temperature and humidity are measured at 1.5 m (4.92 ft.) above the ground and 0.4 m (1.31 ft.) in front of the cabinet. There should be no protection board on the front or back of the cabinet.
- "Short-term" refers to continuous working time that does not exceed 96 hours and an accumulated working time per year that does not exceed 15 days. If the working time exceeds either of these values, it is considered "long-term".

Software Upgrade Paths

Visit [Software Upgrade Paths](#) or contact local Huawei engineers for more detailed information.

For More Information

For more information about the NetEngine 8000 M1A service router, visit <https://e.huawei.com> or contact us in the following ways:

- Global service hotline: <https://e.huawei.com/en/service-hotline>
- Logging in to the Huawei Enterprise technical support web: <https://support.huawei.com/enterprise/>
- Sending an email to the customer service mailbox: support_e@huawei.com

Copyright © Huawei Technologies Co., Ltd. 2025. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions

 HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang Shenzhen 518129
People's Republic of China

Website: www.huawei.com